

■ INTRODUCTION

The CE8510 is a constant frequency, current mode step-down converter. The device integrates a main switch and a synchronous rectifier for high efficiency without an external Schottky diode. It is ideal for powering portable equipment that runs from a single cell Lithium-Ion (Li+) battery. The output voltage can be regulated as low as 0.6V. The CE8510 can also run at 100% duty cycle for low dropout operation, extending battery life in portable system. This device offers two operation modes, PWM control and PFM Mode switching control, which allows a high efficiency over the wider range of the load.

The CE8510 is offered in a low profile 6-pin, SOT package, and is available in an adjustable version.

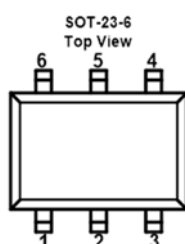
■ FEATURES

- High efficiency :Up to 96%
- 500kHz Constant Frequency Operation
- 2A Output Current
- No Schottky Diode Required
- 4.5V to 18V Input Voltage Range
- Output Voltage as Low as 0.6V
- Slope Compensated Current Mode Control for Excellent Line and Load Transient Response
- Integrated internal compensation
- Stable with Low ESR Ceramic Output Capacitors
- Over Current Protection with Hiccup-Mode
- Thermal Fault Protection
- Inrush Current Limit and Soft Start
- SOT23-6 package
- -40° C to +85° C Temperature Range

■ APPLICATIONS

- Distributed Power Systems
- Digital Set Top Boxes
- Flat Panel Television and Monitors
- Wireless and DSL Modems
- Notebook Computer

■ PIN CONFIGURATION



■ ORDER INFORMATION

CE8510①②

DESIGNATOR	SYMBOL	DESCRIPTION
①	A	Standard
②	E	Package: SOT23-6

Tabel1. Pin Description

PIN NUMBER	PIN NAME	FUNCTION
E		
1	BS	Bootstrap. A capacitor connected between SW and BS pins is required to form a floating supply across the high-side switch driver.
2	V _{SS}	Analog ground pin.
3	FB	Adjustable version feedback input. Connect FB to the center point of the external resistor divider.
4	EN	Drive this pin to a logic-high to enable the IC Drive to a logic-low to disable the IC and enter micro-power shutdown mode.
5	V _{IN}	Power supply Pin
6	SW	Switching Pin

■ ABSOLUTE MAXIMUM RATINGS(Note1)

PARAMETER		SYMBOL	RATINGS	UNITS
Input Voltage		V _{IN}	V _{SS} -0.3~V _{SS} +20	V
EN Voltages		V _{EN}	V _{SS} -0.3~V _{SS} +20	V
FB Voltage			V _{SS} -0.3~ V _{SS} +6	V
SW Voltage			V _{SS} -0.3~V _{IN} +0.5	V
BS Voltage			V _{SW} -0.3~V _{SW} +5	V
Power Dissipation	SOT23-6	P _D	400	mW
Thermal Resistance				
Operating Temperature		T _{opr}	-40~+85	°C
Junction Temperature		T _j	150	°C
Storage Temperature		T _{stg}	-40~+125	°C
Soldering Temperature & Time		T _{solder}	260°C, 10s	
ESD HBM(Human Body Mode)		-	2	kV
ESD MM(Machine Mode)		-	200	V

■ ELECTRICAL CHARACTERISTICS

CE8510 Series ($V_{IN}=V_{EN}=12V$, $V_{OUT}=1.8V$, $T_A = 25^{\circ}C$, unless otherwise noted.)

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
Input Voltage		4.5		18	V
Supply Current in Operation	$V_{EN}=2.0V$, $V_{FB}=1.1V$		0.4	0.6	mA
Supply Current in Shutdown	$V_{EN}=0$ or $EN = GND$		1		μA
Regulated Feedback Voltage VFB	$T_A = 25^{\circ}C$ $4.5V \leq V_{IN} \leq 18V$	0.588	0.600	0.612	V
High-Side Switch On-Resistance			90		m Ω
Low-Side Switch On-Resistance			70		m Ω
High-Side Switch Leakage Current	$V_{EN}=0V$, $V_{SW}=0V$		0	10	μA
Upper Switch Current Limit	Minimum Duty Cycle		2.6		A
Oscillation Frequency			0.5		MHZ
Maximum Duty Cycle	$V_{FB}=0.6V$		92		%
Minimum On-Time			60		ns
Soft-start Time	T_{SS}		4		ms
Thermal Shutdown			160		$^{\circ}C$

Note:

1: Absolute Maximum Ratings are those values beyond which the life of a device may be impaired.

2: T_J is calculated from the ambient temperature T_A and power dissipation P_D according to the following formula: $T_J = T_A + (P_D) \times (170^{\circ}C/W)$.

3: 100% production test at $+25^{\circ}C$. Specifications over the temperature range are guaranteed by design and characterization.

■ TYPICAL APPLICATION CIRCUITS

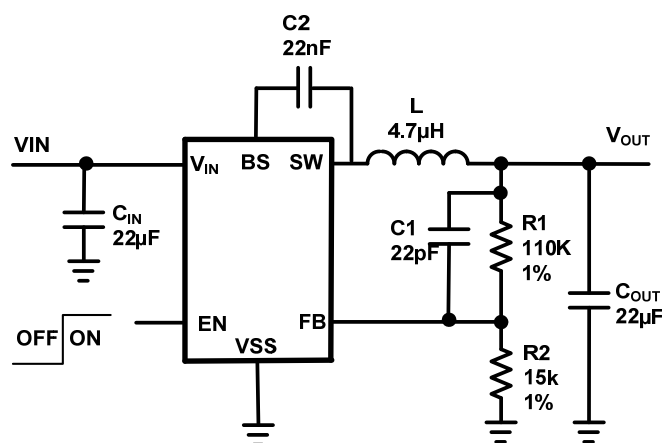
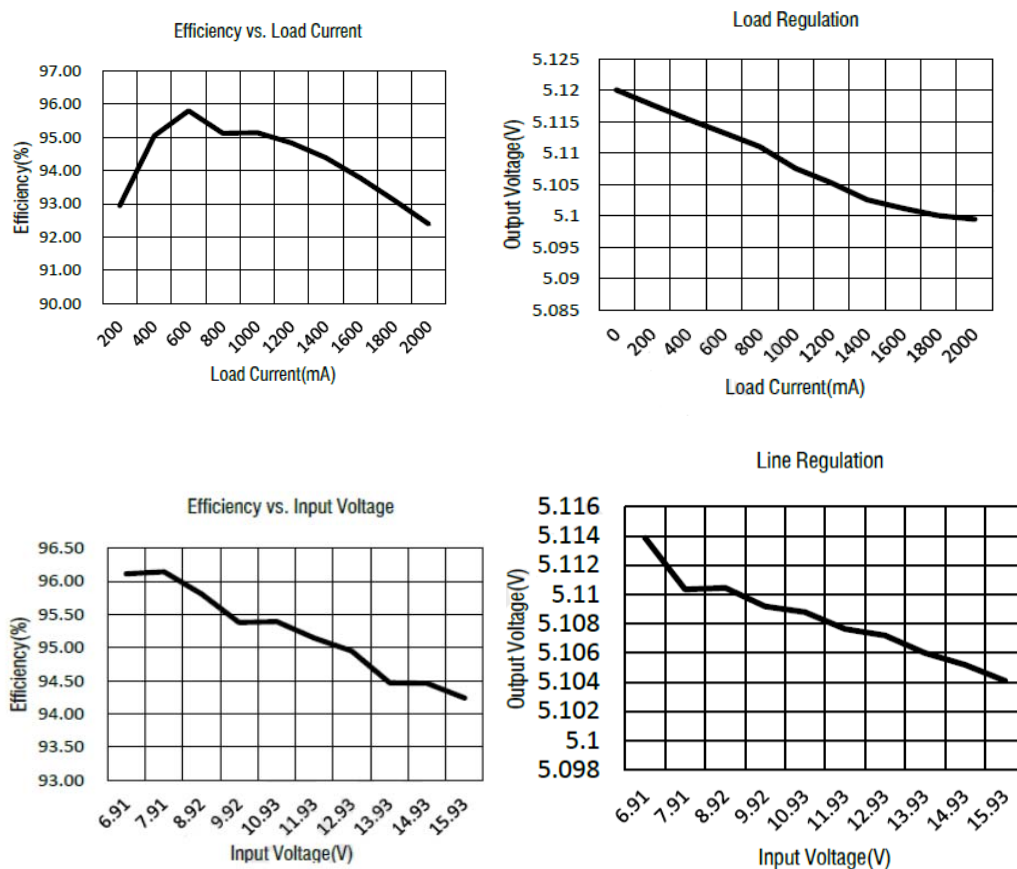


Figure1Basic Application Circuit

$$\text{Note: } V_{\text{OUT}} = V_{\text{FB}} \times \left(1 + \frac{R1}{R2}\right)$$

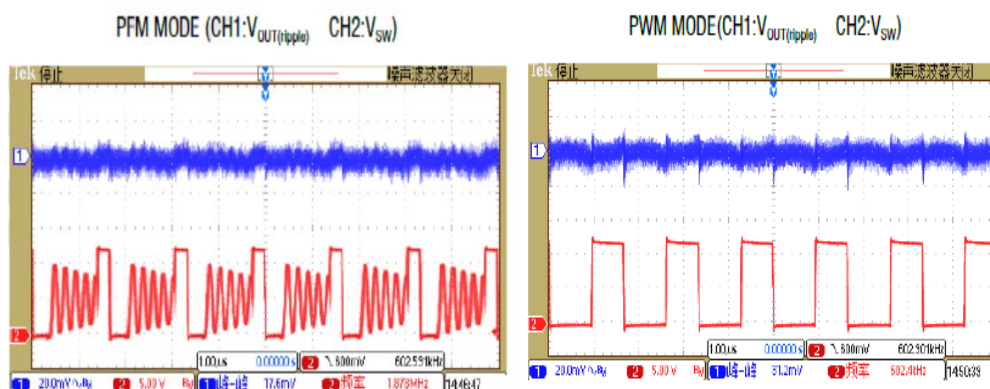
■ TYPICAL PERFORMANCE CHARACTERISTICS

(Test Figure1 above, unless otherwise specified)



■ TYPICAL PERFORMANCE CHARACTERISTICS

(Test Figure1 above, unless otherwise specified)



■ FUNCTIONAL BLOCK DIAGRAM

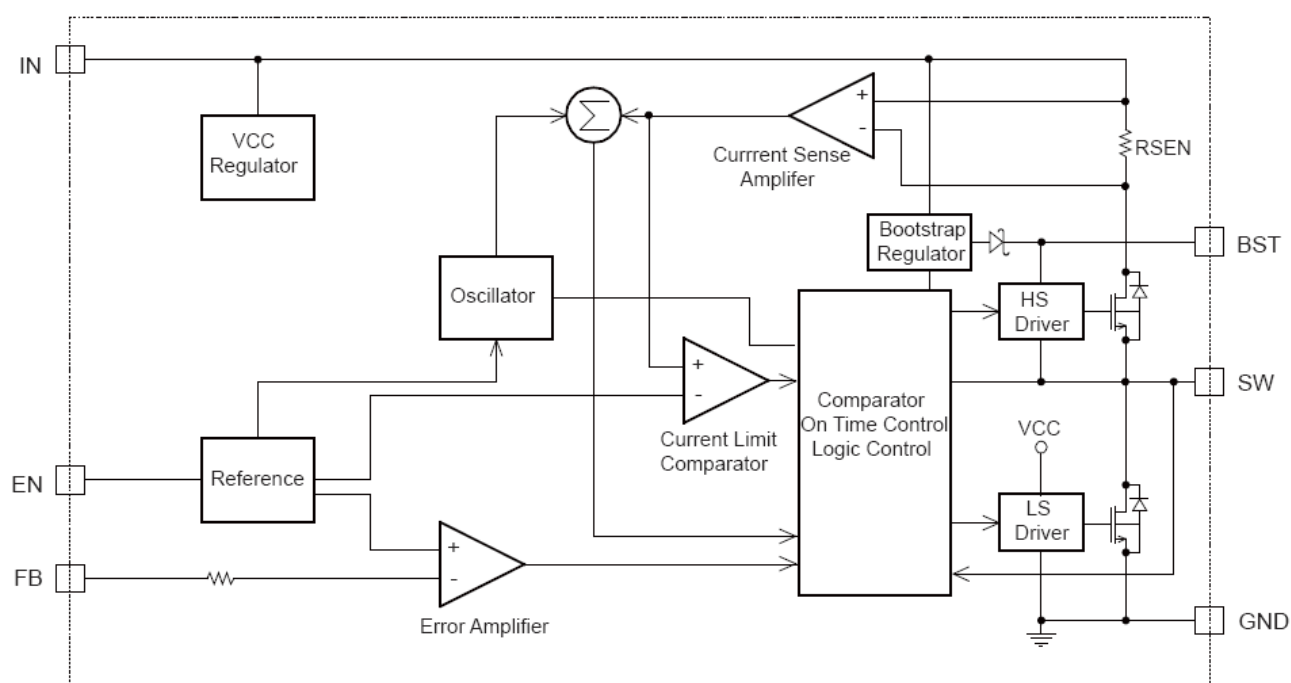
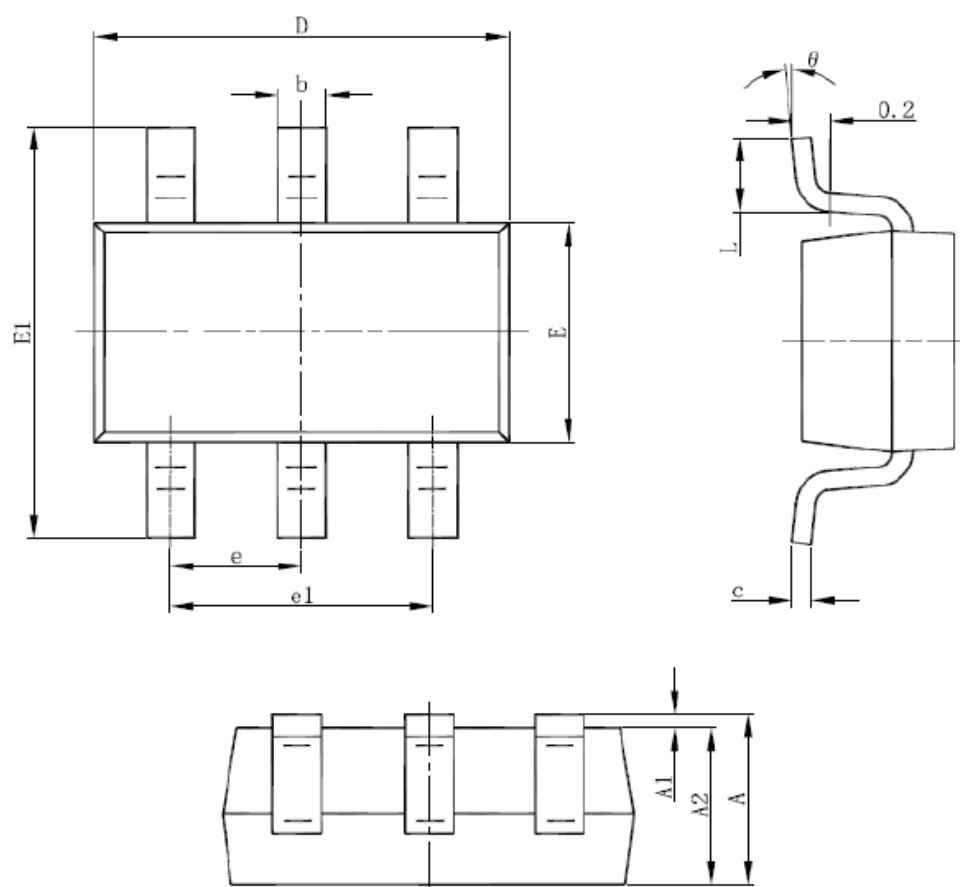


Figure2 Block Diagram

■ PACKAGING INFORMATION

● SOT23-6 Package Outline Dimensions



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

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